

Video RAG Retrieval Project

<https://github.com/aiyshwary/Video-RAG-Retrieval>

OVERVIEW

A semantic video search system that extracts frames, generates scene captions using ViT-GPT2, and embeds them using SentenceTransformers and OpenCLIP into a dual FAISS index, enabling natural language queries to retrieve relevant video segments with timestamps.

IMPACT

Built a Visual RAG system that makes unstructured video content semantically searchable: a text query returns relevant video timestamps by matching against dual text and image embedding indexes via cosine similarity.

TECHNICAL WALKTHROUGH

I built a Visual Retrieval-Augmented Generation (RAG) system that retrieves relevant video scenes based on user queries by combining vision models and vector similarity search.

Searching videos using text is hard because video content is unstructured. Traditional keyword search is limited to finding keywords happening inside a video. My project solves this by converting both video scenes and user queries into vector embeddings and retrieving relevant video segments.

- i) Scene understanding using vision models
- ii) Vector storage for text and image embeddings
- iii) Similarity-based retrieval using VDMS
- iv) Video → Scene Description + Frames → Embeddings → Vector DB → Retrieval → Video Clips

The input to the system is one or more videos. Each video is processed to extract: Scene-level textual descriptions using models like Video-LLaMA. Key video frames along with metadata like frame number, FPS, and timestamps.

Vision-language models are used to understand the video content. These models analyze video scene descriptions that summarize what's happening in each scene. For example, a scene might be described as 'a person picking up an item'.

To support efficient retrieval, two types of embeddings are stored in separate vector stores:

- i) Text embeddings: Model: SentenceTransformer – all-MiniLM-L6-V2 / L12-V2. Used for scene descriptions. Converts textual descriptions into dense vectors that capture semantic meaning.
- ii) Image embeddings: Model: OpenCLIP – ViT-g-14. Used for video frames. This helps capture visual scene context directly from frames.

Both text and image embeddings are stored in VDMS, which allows fast vector similarity search. Separate vector stores for text and image embeddings but are linked using video metadata.

- i) Stored metadata includes: Video name, Scene ID, Frame number, Timestamp

When a user enters a text prompt: Prompt is converted into a text embedding. Vector similarity search is performed over text embeddings and (optionally) image embeddings. The system retrieves the most relevant scenes and clips.

Scene description and exact time range to jump to in the video. Text embeddings capture semantic meaning, while image embeddings capture visual features. Using both improves retrieval accuracy because some queries are better answered by visual context.

- i) 'a red car' → image embeddings
- ii) 'a tense conversation' → text embeddings
- iii) Efficient vector search using cosine similarity
- iv) Decoupling embeddings for scalability
- v) Model choice based on speed vs accuracy
- vi) Frame sampling using FPS to avoid redundancy

This approach makes video content searchable, explainable, and scalable, and it can be extended to support enterprise video search.

A Visual RAG system enables semantic search over videos. It uses vision models to generate scene descriptions, converts them into text and image embeddings using SentenceTransformers and OpenCLIP, stores the embeddings, and retrieves video segments based on user queries.

KEY DETAILS

1. Extracts frames at a configurable FPS and generates natural-language scene captions using ViT-GPT
2. Computes text embeddings with SentenceTransformers (MiniLM) and image embeddings with OpenCLIP
3. Stores embeddings and metadata in two parallel FAISS indexes supporting independent text or image search
4. Query interface accepts a natural language string and returns top-k matching scenes with frame references
5. Modular architecture (preprocessing, embeddings, storage, retrieval) designed for extension to production
6. FAISS indexes use IndexFlatIP with L2-normalized vectors for cosine-similarity search; OpenCLIP default is LAION-2B with automatic GPU/CPU selection.
7. Includes a pytest suite and a self-contained demo that synthesizes a 30-frame test video with OpenCV pipeline without needing a real video file.